

The Geometry of Time

How Topology Dictates Decay in Knotted Vortex Filaments

In the realm of quantum hydrodynamics, the shape of a knot is destiny. This presentation introduces a new framework establishing that “Writhe”—geometric complexity—acts as the primary variable controlling the lifespan of vortex filaments.

Based on the research of O. Iskandarani (2026)

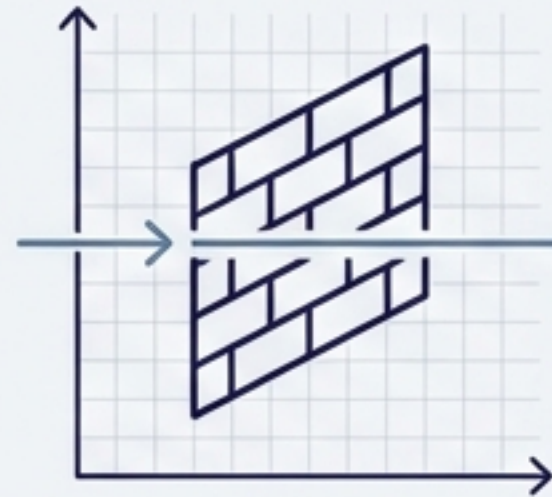


Executive Summary: Complexity is a Tax on Survival



High Complexity (High Writhe)

Intrinsic geometric phase lags.



Low Reynolds Barrier

Re_{α} scales as Wr^{-2}



Rapid Decay

Short Lifetime



High Reynolds Barrier



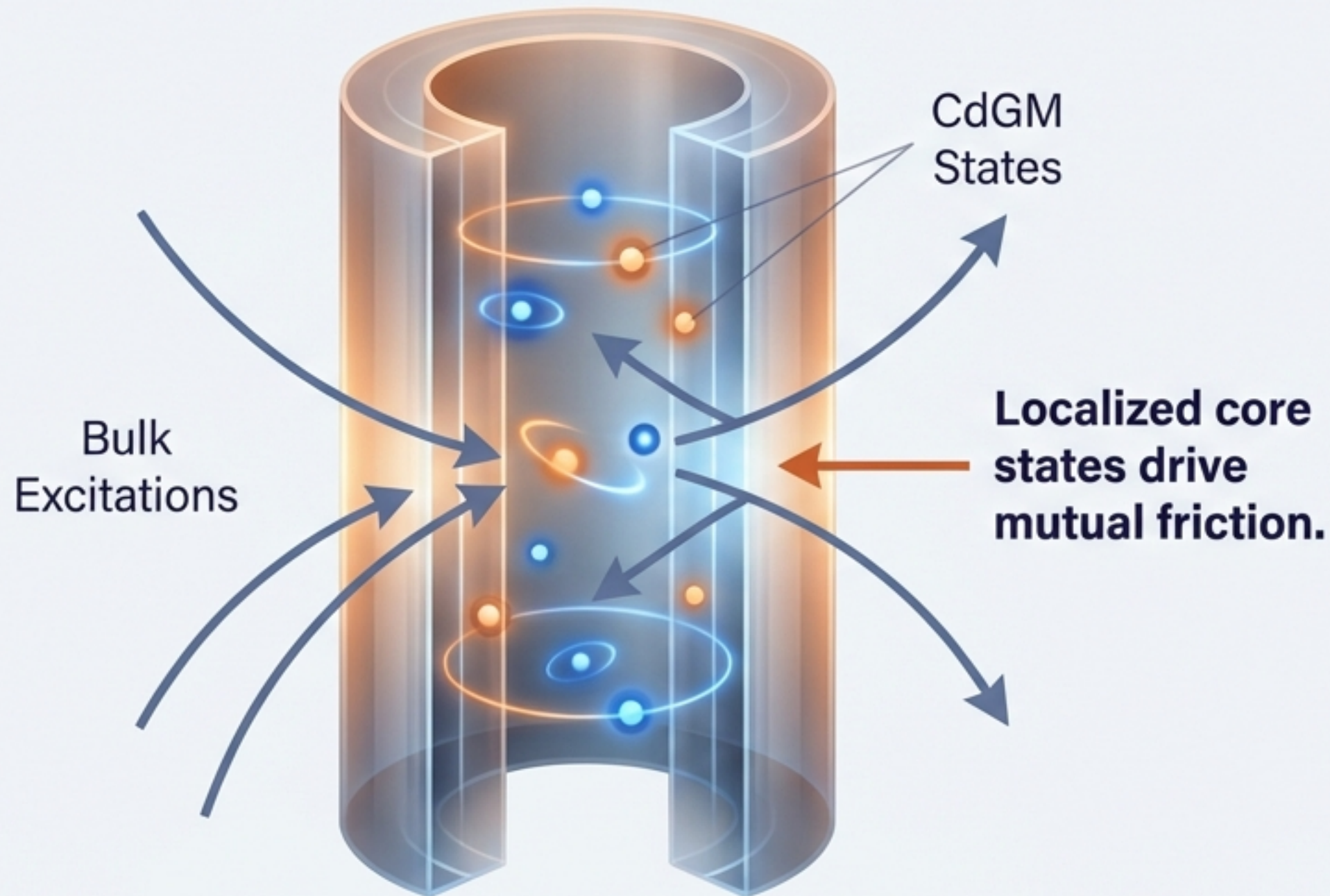
Ring (Simple) -> High Reynolds Barrier -> Long Stability

Key Takeaways

- **The Mechanism:** Non-planar geometries introduce phase lags, reinterpreting friction as geometry.
- **The Barrier:** Stability scales inversely with complexity.
- **The Result:** A predictable hierarchy of lifetimes independent of particle identity.

The Spark: Vortex Dynamics & Core States

Contextualizing the Grani et al. (2025) Breakthrough



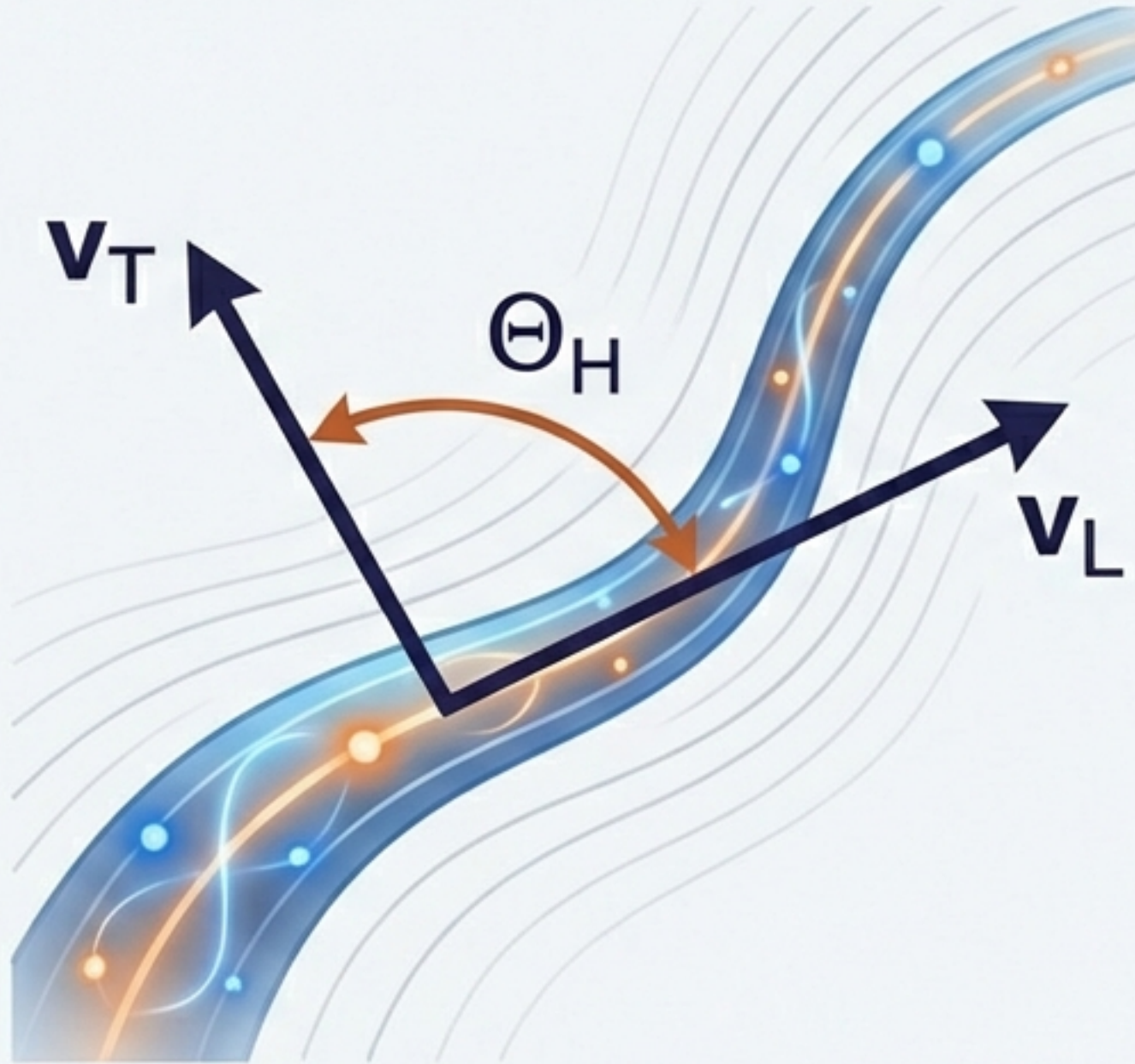
- **The Shift**

2025 experiments established the Hall Angle (Θ_H) as a key observable linked to internal states.

- **The Pivot**

If internal states matter, does the macroscopic *shape* of the knot also change the physics?

Reinterpreting Friction as Geometry



$$\tan(\Theta_H) = \frac{1 - \alpha'}{\alpha}$$

- α (Dissipative): Standard friction.
- α' (Reactive): Traditionally transport, now reinterpreted as **Geometric Phase Lag**.

In non-planar filaments, torsion produces a geometric phase shift ($\Delta \alpha \alpha'_{\text{geom}}$) that survives even in the low-temperature limit.

The Protagonist: Writhe (Wr)

The DNA of Geometric Complexity



Unknot, $Wr = 0$

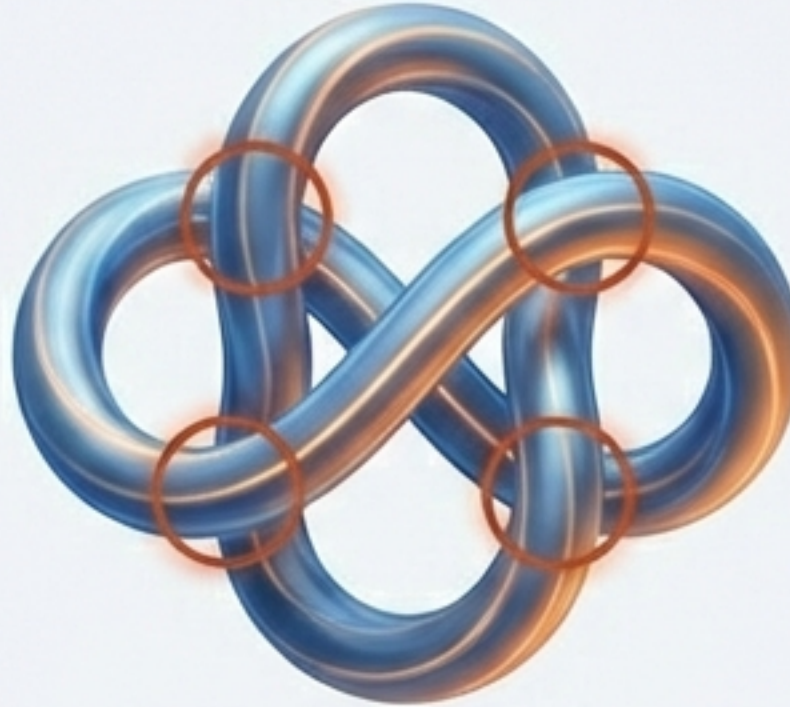
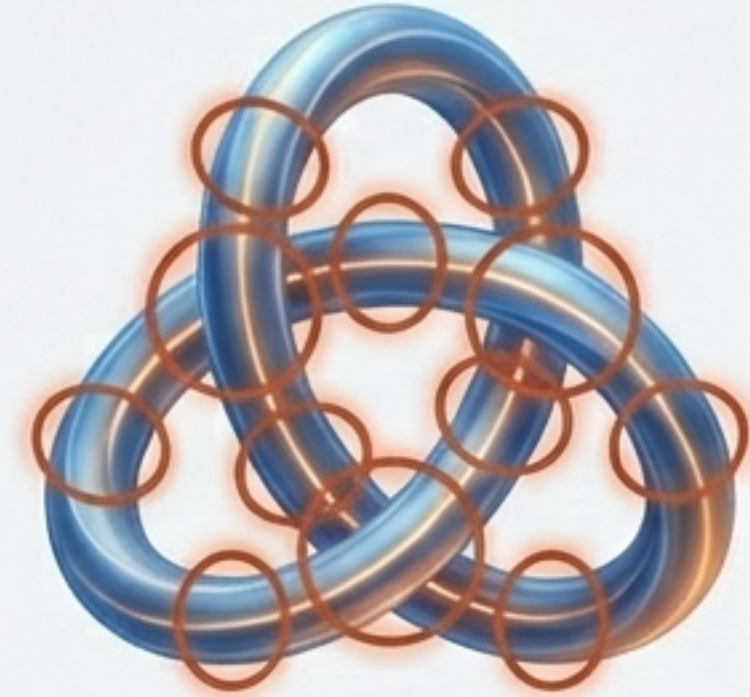


Figure-8, Medium Wr

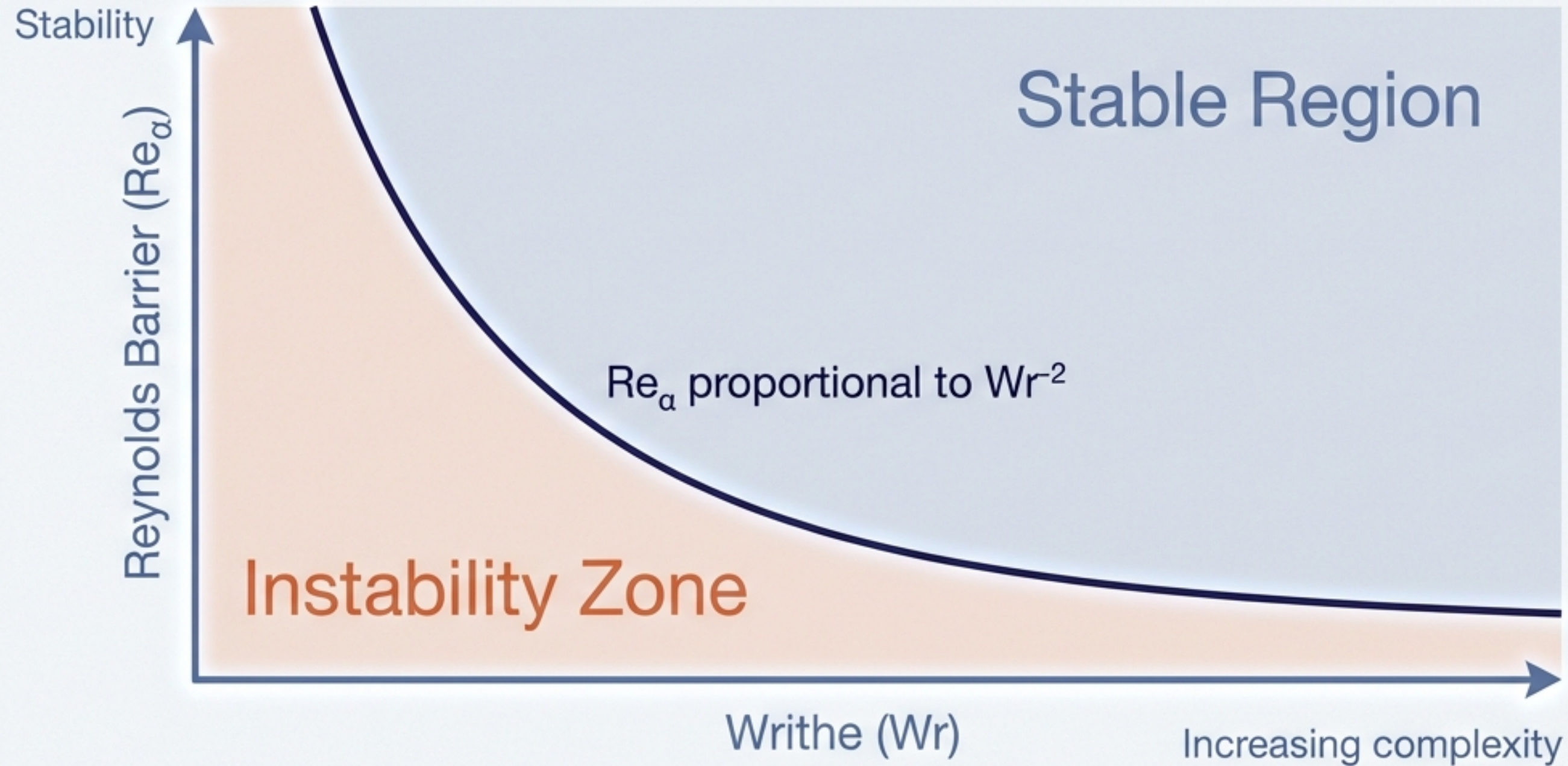


Trefoil, High Wr

$$Wr = \frac{1}{4\pi} \iint \frac{(\mathbf{r}_1 - \mathbf{r}_2) \cdot (d\mathbf{r}_1 \times d\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3}$$

Writhe measures non-planarity and spatial self-entanglement. It acts as the Geometric Control Parameter (χ_K). High Writhe = High self-induced velocity fluctuations.

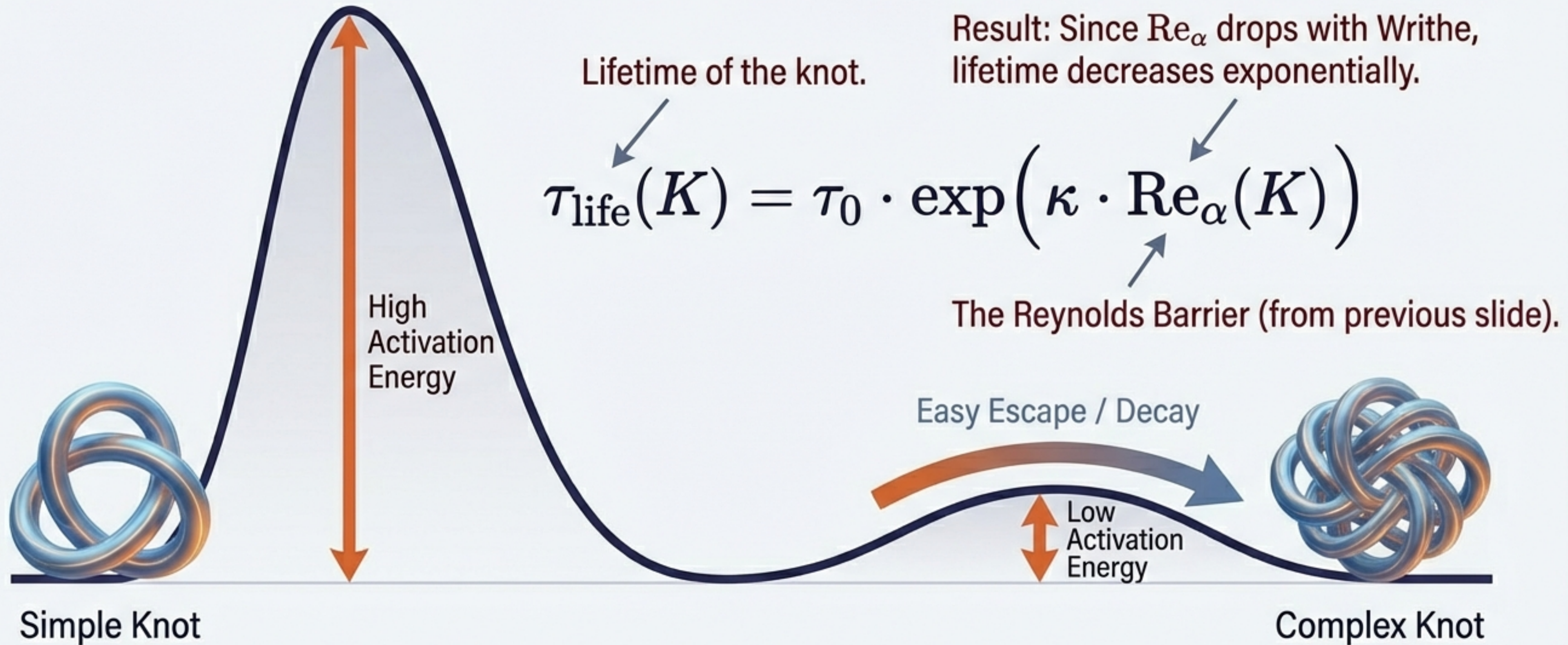
The Topological Reynolds Barrier



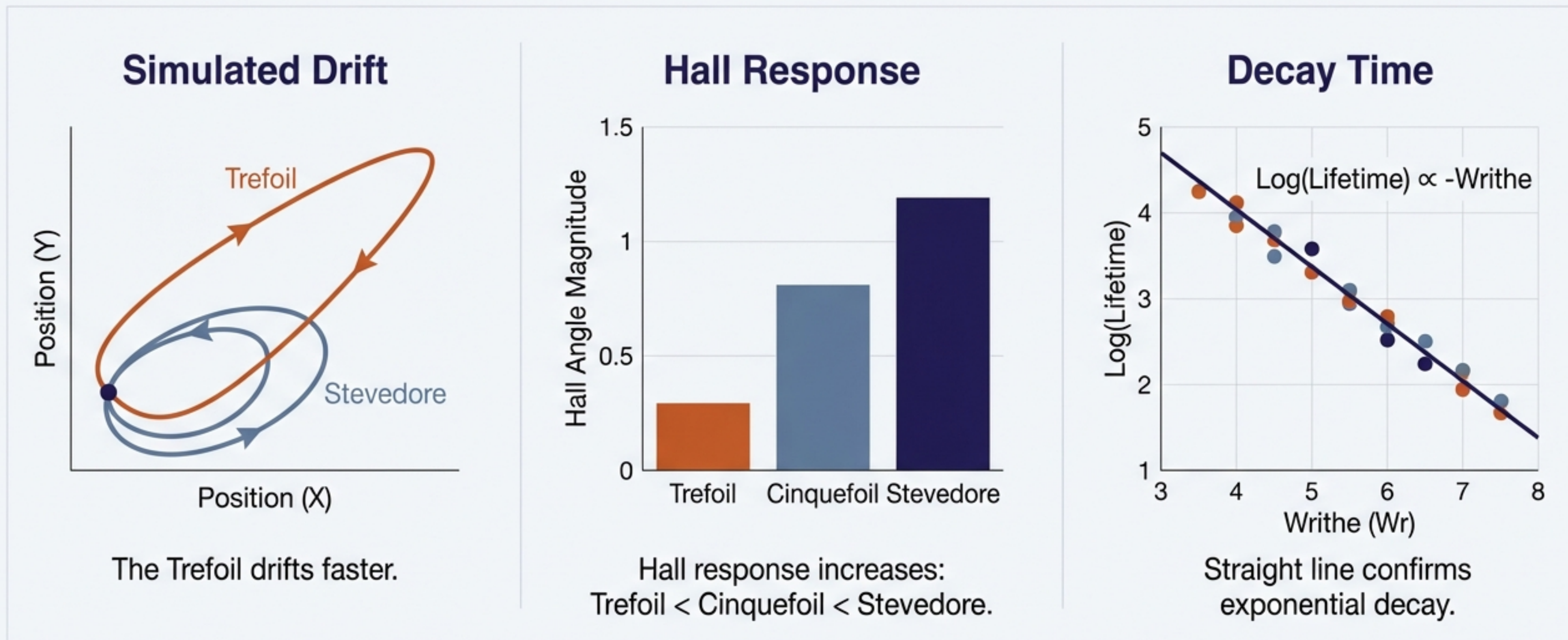
We introduce a dimensionless stability parameter. Filaments with larger Writhe experience enhanced self-scattering. Complexity is a burden: the more complex the knot, the lower the barrier to instability.

The Law of Decay: Lifetime Scaling

Decay follows a strict topological hierarchy



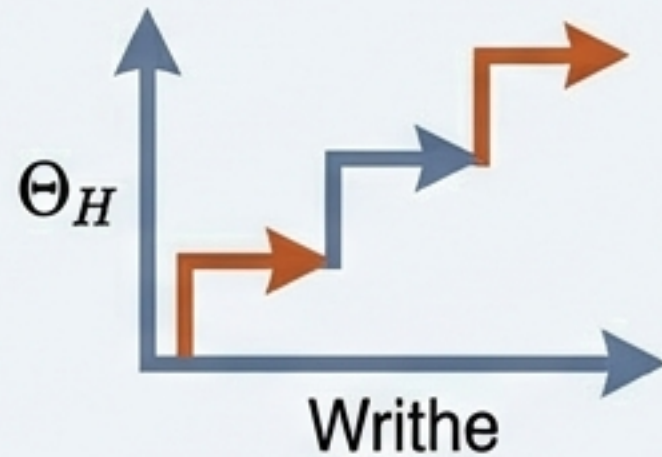
Numerical Evidence Confirms the Writhe-Lifetime Link



Simulations tracked transverse drift and time-to-reconnection. Results confirm that Hall response increases with Writhe, while lifetime decreases exponentially, independent of microscopic friction models.

Testable Predictions & Falsifiability

✓ Discrete Hall Plateaus



Distinct topologies must show distinct Hall angles. The difference is proportional to the difference in Writhe.

$$\Theta_H(K_1) - \Theta_H(K_2) \propto \text{Wr}(K_1) - \text{Wr}(K_2)$$

✓ Universal Lifetime Ratios



The ratio of lifetimes between two knot types is parameter-free. It depends only on their geometry.

$$\frac{\tau(K_1)}{\tau(K_2)} = \exp \left[\kappa \cdot (\text{Wr}(K_2)^{-2} - \text{Wr}(K_1)^{-2}) \right]$$

Any deviation from these scaling laws would falsify the Reynolds-barrier mechanism.

A Universal Organizing Principle

Beyond Fermi Gases: Analogies in Nonlinear Continua



Superfluids



Magnetic Flux Tubes



Optical Vortices

While the model implies no microscopic particle identity, **the link Writhe and Stability** appears universal. **Topological Hall response** serves as a measurable proxy for internal relaxation across these diverse systems.



DNA Supercoiling

Limitations and Validity

Hydrodynamic Effective Theory

This is a **macroscopic description**. It does not claim to describe the **fundamental microscopic particle structure**. It is complementary to field-theoretic descriptions.

Approximations

1. **Slender-Filament Approximation**: Assumes thin vortex lines.
2. **Phenomenological Reconnection**: Modeled effectively rather than microscopically.

Neglected Effects

Thermal and quantum fluctuations beyond **effective friction** are **ignored**. The model focuses purely on the geometry-driven dynamics.

Geometry is the Architect of Dynamics

We have established that the Hall dynamics of knotted filaments encode a hierarchy of relaxation times.

A Reynolds-like barrier, scaling inversely with Writhe, governs the life and death of topological excitations.

In the complex environment of quantum fluids, shape is destiny.

